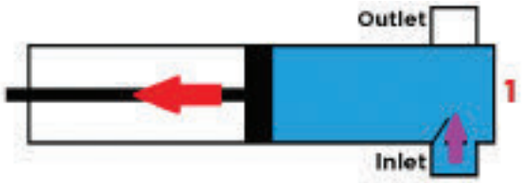
Working

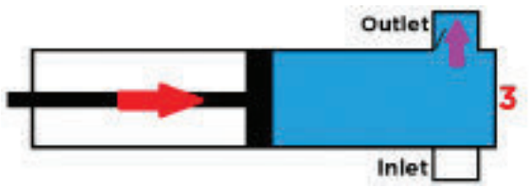
Action1: The piston is pulled back. This action increases the volume of the cavity (cylinder). As the cavity volume expands, fluid is drawn in through the inlet to fill the expanded cavity (Suction stroke).



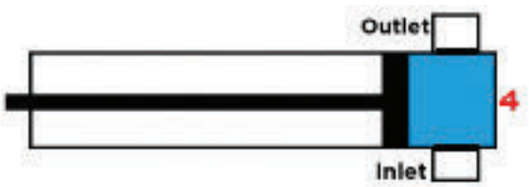
Action 2: The piston has reached its maximum displacement. Since it is not moving into or out of the cavity, fluid is not following through the inlet or the outlet.



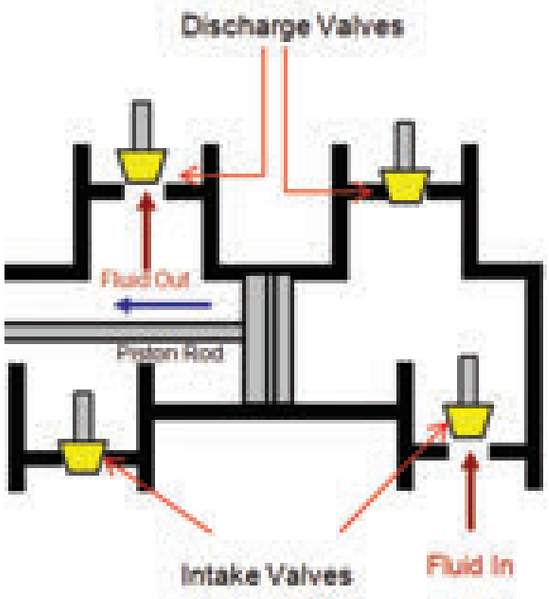
Action 3: The piston is then pushed back into the cavity. During this process, the piston applies enough pressure in the outlet of the pump. This pressure pushes the fluid from inside the cavity through outlet of the pump. (Delivery stroke)



Action 4: The piston reaches its maximum extension into the cavity. Here the volume of the cavity at a minimum and fluid is not flowing through the inlet or the outlet. The next action repeats the process starting again with action1.



7.6.5 Double acting reciprocating pump



Working

It is similar to single acting reciprocating pump. It has two suction and two delivery pipes and valves on either side of the piston connected to the cylinder

When piston moves, water is collected on one side and discharged on other side, so the water is discharged continuously.

There are two suction strokes and delivery strokes during each revolution of the crank. Thus, water is supplied uniformly in the delivery pipe and the quantity of discharge is double that of single acting reciprocating pump.

7.6.6 Centrifugal Pump

A hydraulic machine which converts the mechanical energy into the pressure energy by means of centrifugal force, used to pump a fluid is called centrifugal pump.

The following are the main parts of centrifugal pump.

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